Question-5.13.48

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Question

If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of equal degree, then which one is correct among the followings?

1. $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
2. $\mathbf{A} + \mathbf{B} = \mathbf{A} - \mathbf{B}$
3. $\mathbf{A} - \mathbf{B} = \mathbf{B} - \mathbf{A}$
4. $\mathbf{AB} = \mathbf{BA}$

Solution

For any two matrices $\mathbf{A}$ and $\mathbf{B}$ of the same order, addition is commutative
Therefore, $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.
let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 3 & 6 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 5 & 3 \\ 8 & 2 \end{pmatrix} \tag{1}$$

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} 2+5 & 3+1 \\ 3+8 & 6+2 \end{pmatrix} \tag{2}$$

$$= \begin{pmatrix} 7 & 4 \\ 11 & 8 \end{pmatrix} \tag{3}$$

$$\mathbf{B} + \mathbf{A} = \begin{pmatrix} 5+2 & 1+3 \\ 8+3 & 2+6 \end{pmatrix} \tag{4}$$

$$= \begin{pmatrix} 7 & 4 \\ 11 & 8 \end{pmatrix} \tag{5}$$

Therefore, $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.

C-Code

```c
// code.c
#include <stdio.h>

void get_points(double *A, double *B) {
    // 2D points A: (1, 2), B: (3, 4)
    A[0] = 1; A[1] = 2;
    B[0] = 3; B[1] = 4;
}
```

Python-Code

```python
# Code by GVV Sharma
# Modified for Problem Solution
# Released under GNU GPL
# Calculating area enclosed between curves
# call.py
import ctypes
import numpy as np

lib = ctypes.CDLL('./code.so')
```

Python-Code

```
get_points = lib.get_points
get_points.argtypes = [np.ctypeslib.ndpointer(dtype=np.float64,
    shape=(2,)),
                       np.ctypeslib.ndpointer(dtype=np.float64,
                           shape=(2,))]
get_points.restype = None

def get_points_from_c():
    A = np.zeros(2, dtype=np.float64)
    B = np.zeros(2, dtype=np.float64)
    get_points(A, B)
    return A, B
```